Ved Chauhan

Software Engineer — Systems, Full-Stack & Developer Tools | **Ahmedabad, India**

vedchauhan2107@gmail.com | github.com/VED2107 | linkedin.com/in/ved-chauhan2107 | ved.exe.snowbros.me | snowbros.me

SUMMARY

Computer Engineering student and founder of SNOWBROS, shipping tested software end to end: a Rust static-analysis toolchain on npm and crates.io, a Flutter file-transfer app for Android and Windows, a PostgreSQL accounting product, a 7,044-job docking pipeline for a client, and retail software in a paying client's store.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Dart, Python, SQL
Frontend & Apps: React 19, Next.js (App Router), Flutter, Tailwind CSS, Tauri
Backend & Data: Node.js, REST APIs, PostgreSQL, Supabase, Row-Level Security, database design, authorization
Systems & Tooling: Tree-sitter, static analysis, LSP, TCP/HTTP, job queues, Docker, RDKit, AutoDock Vina
Testing & DevOps: Vitest, Playwright, pytest, cargo test, dart test, GitHub Actions CI/CD, release automation
Foundations: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

EXPERIENCE

Founder & Software Engineer — SNOWBROS, independent software studio — Feb 2026 – Present

- Own client and open-source products end to end: schema design, code, tests, CI/CD and releases.
- Delivered **Atelier**, a retail operating system for a paying fashion boutique: 35-table PostgreSQL schema with Row-Level Security, Razorpay UPI settled only by an HMAC-verified webhook, and an offline-first Tauri/SQLite till; 268 Vitest tests plus Playwright E2E in CI.

PROJECTS

Snowbros Atlas — Open-Source Static Analysis Toolchain | github.com/snowbros-labs/atlas — Jul 2026 – Present

Rust, Tree-sitter, petgraph, tower-lsp, TypeScript, GitHub Actions

- Architected a 19-crate Rust workspace (~17k LOC, 340 tests) that lowers JavaScript/TypeScript and Python through Tree-sitter into one shared IR and runs 20+ rules over a project-wide import/symbol graph.
- Engineered an incremental xxh3 cache with byte-identical warm output: axios (431 files) analyzes in ~230 ms cold / ~76 ms warm; a 500-file repo re-analyzes in ~34 ms after a one-file change.
- Shipped a CLI, LSP server, VS Code extension and SARIF output; 5 GitHub Actions workflows release v0.4.0 to npm, crates.io, Homebrew and x64/arm64 binaries for three OSes.

SwiftDrop — Cross-Platform LAN File Transfer | github.com/VED2107/swiftdrop — Sep 2026 – Present

TypeScript, Node.js, Flutter, Dart, React, Vitest, Playwright

- Engineered a resumable, parallel transfer engine (per-MiB xxh64 checks, bitmap resume, adaptive concurrency), and ported it to a pure-Dart core shared by the Flutter Android and Windows apps.
- Raised loopback throughput 232→314 MB/s (10k × 50 KB) and 66→111 MB/s (1k × 10 KB) with contiguous batch frames, measured by a two-process benchmark harness; shipped v1.1.0 installer and APK.

AMR Drug Repurposing — Computational Screening Platform (Client Project) — Sep 2026 – Present

Python, RDKit, scikit-learn, AutoDock Vina, PostgreSQL, Supabase, Docker, Next.js

- Built a distributed docking campaign for 1,761 approved medicines × 4 bacterial targets (7,044 AutoDock Vina jobs): a Postgres SKIP LOCKED queue with heartbeat leases, backoff retries and SHA-256-checked inputs.
- Engineered ChEMBL/FDA/ClinicalTrials.gov ingestion, scaffold-split ML models and a Next.js results site; fixed train/test scaffold leakage. Results are framed as computational predictions, not clinical evidence.

Accounic — Multi-Currency Accounting App | github.com/VED2107/accounic — Aug 2026 – Present

PostgreSQL, Supabase, Next.js, TypeScript, Flutter, GitHub Actions

- Cut dashboard load from 98.2 s to 1.2 s (82×) at 20,000 entries by moving a per-entry FIFO settlement allocator into a per-person CTE, with per-currency totals asserted byte-for-byte unchanged.
- Shipped v1.16.1 to web, Android and Windows from one SQL accounting engine with RLS tenant isolation; 5 CI workflows run 15 SQL suites against a throwaway Postgres plus web and Flutter tests.

EDUCATION

B.Tech in Computer Engineering — U.V. Patel College of Engineering, Mehsana — Expected May 2028

CGPA: 7.5 / 10.0 | IEEE Student Member